



THE IV LEAGUE

A fully autonomous, self-improving ODTE options trading system — eight cooperating AI agents and deterministic risk logic, ranking, sizing, executing, and reviewing real trades against a live brokerage account every trading day.

4 live trading days

Aug 31 – Sep 3, 2026

13 real trades

paper account, real fills

101 / 101 tests

TDD throughout

Alpaca

Trading API · MCP · CLI

One universe, two strategies, one rule

Every morning the system ranks 8 underlyings by option-chain IV skew and splits them by conviction: the calm names get sold as premium, the volatile ones get read for direction.

TOP 3 BY IV SKEW

Premium-Selling

Sell the ~15Δ ODTE put, cash-secured. Pure mechanics — no directional bet, just collecting skew. 50% take-profit / 3× stop-loss / end-of-day close.

BOTTOM 5, ANALYST-READ

Directional

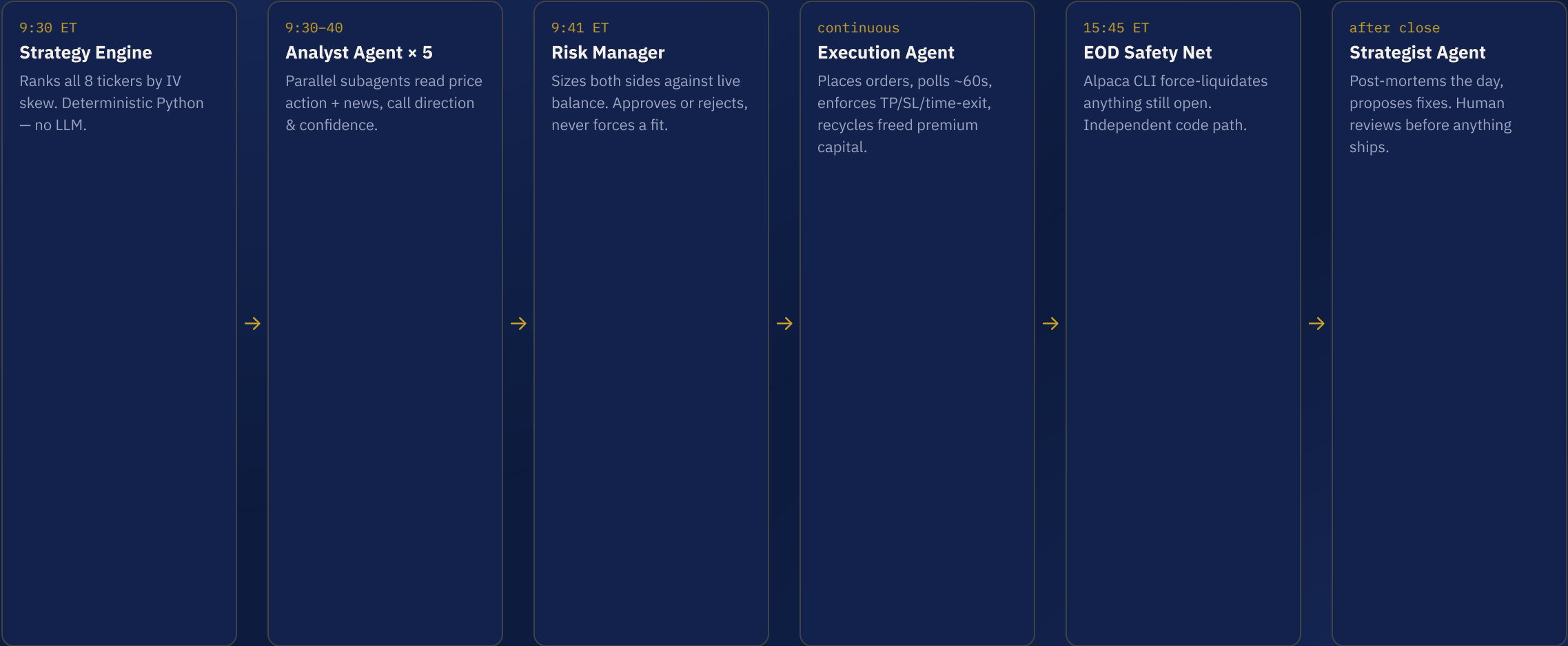
An Analyst agent reads first-session price action + news, calls Bullish / Bearish / Undecided. Confidence-gated selection buys an ATM ODTE call or put.

GOVERNING PRINCIPLE (SPEC §30)

AI agents supply **judgment** — reading news, price action, and proposing process fixes. They never touch risk sizing or order placement: that stays deterministic Python, fast and unambiguous, with the AI's authority capped at "recommend," never "execute."

The daily pipeline

Six stages, market open to end-of-day, each owned by one agent or deterministic script — cron-scheduled, no manual step required.



Runtime: Claude Code subagents **Data/Execution:** Alpaca Trading API (direct SDK) **Agent tools:** Alpaca MCP server **Ops/Safety net:** Alpaca CLI

How a candidate becomes a trade

Ranking, selection, and sizing are each one deterministic step — the only judgment call in the whole pipeline is the Analyst's directional read.

$$\text{IV Skew} = 15\Delta \text{ Put IV} - \text{ATM IV} \quad \cdot \quad \text{ATM IV} = (\text{ATM Call IV} + \text{ATM Put IV}) / 2$$

PREMIUM-SELLING

- 1 Top 3 tickers by IV skew, descending.
- 2 Sell the ~15Δ put, cash-secured — full strike × 100 reserved, no protective leg.
- 3 Budget split equally across candidates, leftover pooled to rescue anyone who couldn't afford their own share — never forced.
- 4 Exit at 50% take-profit, 3× stop-loss (real standing order), or EOD close.

DIRECTIONAL

- 1 Analyst reads first-session bars + 24h news per remaining 5 tickers.
- 2 Calls **BULLISH** / **BEARISH** / **UNDECIDED** with a confidence score.
- 3 Deterministic filter: drop UNDECIDED, drop confidence < 60, keep top 3.
- 4 Buy ATM ODTE call/put, sized at 0.5% of balance per name (cap 1.5%). No TP/SL — 30-minute time exit only, max loss capped at premium paid.

Four real trading days, real fills

Paper account, real market data, real order fills and stops at Alpaca — not a backtest.

ACCOUNT

\$94,335

from \$100,000 starting

PREMIUM-SELLING

+\$17

6 trades · 83% win rate

DIRECTIONAL

-\$5,676

7 trades · 14% win rate

TREND

↓91%

daily loss, day 1 → day 4



Evidence-driven tuning, human-approved

A Strategist agent post-mortems every trading day and proposes fixes; a human reviews and approves each one before it ships. Nothing changes live unreviewed.

Aug 31	Concentration cap removed Caught live: a per-name cap was rejecting every premium candidate on a real run. Fixed same-day.	STRATEGIST
Aug 31	Ranking-consistency bug fixed Two independently-timed fetches let the same ticker appear on both sides. Now one shared ranking snapshot for the whole run.	STRATEGIST
Sep 2	Directional risk cut, 3 days of evidence + external research Confidence bar 55→60, exit changed from fixed 2:30pm to 30-min hold, allocation halved 1%→0.5%. Sourced against academic ODTE research.	HUMAN
Sep 2	Position recycling built Premium capital freed by an early close now re-enters a fresh candidate instead of sitting idle for the rest of the day.	HUMAN
Sep 2	1DTE fallback for no-ODTE days A ticker with no same-day chain now falls back to the next trading day's contract instead of skipping — still force-closed same-day.	HUMAN

Built to fail safely, not just to work

Real production incidents happened during this build — what mattered was how the system responded.

Fail-closed by design

Sep 3: the automated pipeline crashed on a headless auth error before ranking even started. **Zero bad orders were placed** — `set -e` aborted the run before execution could ever fire off a broken decision.

Root-caused, not patched

ODTE contracts return null greeks from the broker by design (T=0 division). Fixed with a locally-built, tested Black-Scholes solver validated against real market data before going live.

Test-driven throughout

101 / 101 tests passing. Every deterministic component — sizing, exit logic, position building — was written test-first and re-verified live against real Alpaca data.

Independent safety net

A separate end-of-day liquidation script, using the Alpaca CLI instead of the main SDK path, force-closes anything still open — shares no code with whatever might be broken upstream.

What four real days taught the system

What's working

- ▶ Premium-selling: mechanical, IV-skew-driven, **83% win rate** and consistently near-breakeven-to-positive across every single day.
- ▶ The HITL review loop caught and fixed 2 real bugs live, on the day they happened, not weeks later.
- ▶ Every design deviation from the automated path — including today's late manual recovery run — stays fully auditable in the decision files.

What's still being tuned

- ▶ Directional has been the persistent drag (14% win rate) — sizing cut in half and hold time shortened Sep 2, still being watched.
- ▶ Open question: is a naked long the right instrument here, or a defined-risk spread?
- ▶ Per-trade IV logging (isolate theta vs. vega/slippage) not yet built.